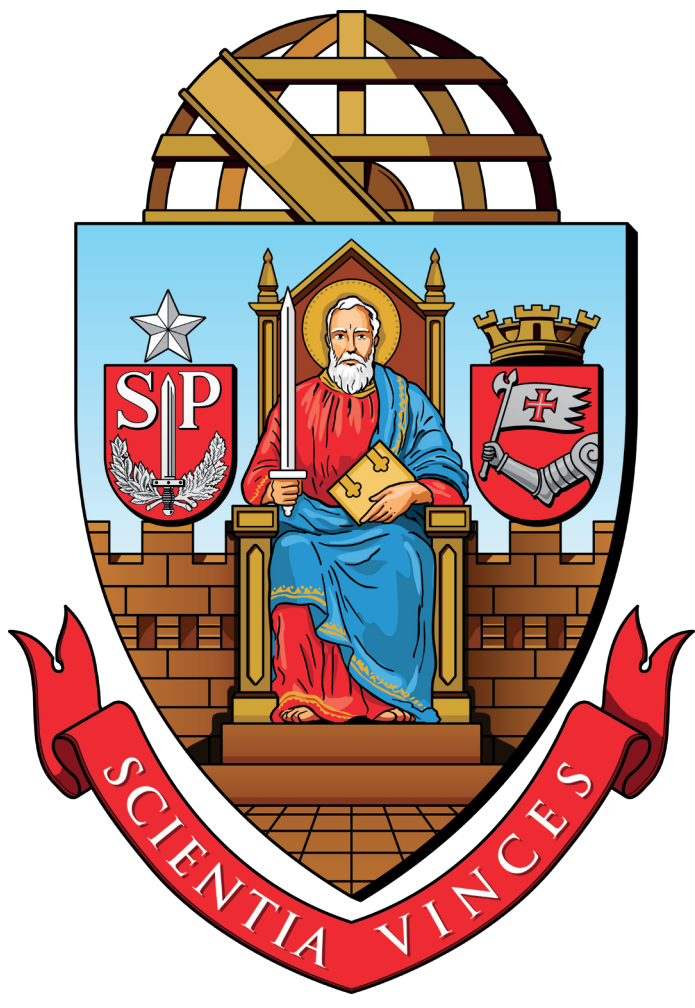




Mining Biosynthetic Gene Clusters in *Helicobacter pylori* Genomes Associated with Premalignant and Malignant Gastric Phenotypes: PRELIMINARY RESULTS AND NEXT STEPS

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Background

- Biosynthetic gene clusters (BGCs) encode microbial secondary metabolites — several with documented antitumor activity in other bacterial genera.
- No prior study has examined *H. pylori* BGCs in the context of gastric carcinogenesis; existing gastric-microbiome-cancer literature is taxonomic/functional, not biosynthetic.
- Pilot study (undergraduate thesis, 2025): 2 metagenome-assembled *H. pylori* genomes from high-risk gastric cancer biopsies → 28 putative BGCs recovered despite only ~2.2% of reads being bacterial, including *de novo* lipogenesis genes (acetyl-CoA carboxylase, β -ketoacyl-ACP synthase, 3-oxoacyl-ACP reductase) and capsular-polysaccharide / tRNA-modification genes with documented tumor-progression parallels.

This Master's stage scales up using public preassembled genomes instead of *de novo* metagenomic assembly, which is limited at scale by low bacterial coverage in tissue.

Objectives

- Mine BGCs across 554 public *H. pylori* genomes (179 from gastric cancer samples vs. 375 from precancerous cases: chronic gastritis / intestinal metaplasia).
- Delineate Gene Cluster Families (GCFs) with a tool-agnostic clustering method spanning antiSMASH, DeepBGC, and GECCO output.
- Test BGC/GCF repertoires for statistical association with gastric cancer phenotype, then validate for biological consistency.
- Correlate phylogeny with BGC/GCF repertoire, search for a combined malignancy signature, and plan functional validation in an animal model.

Methods

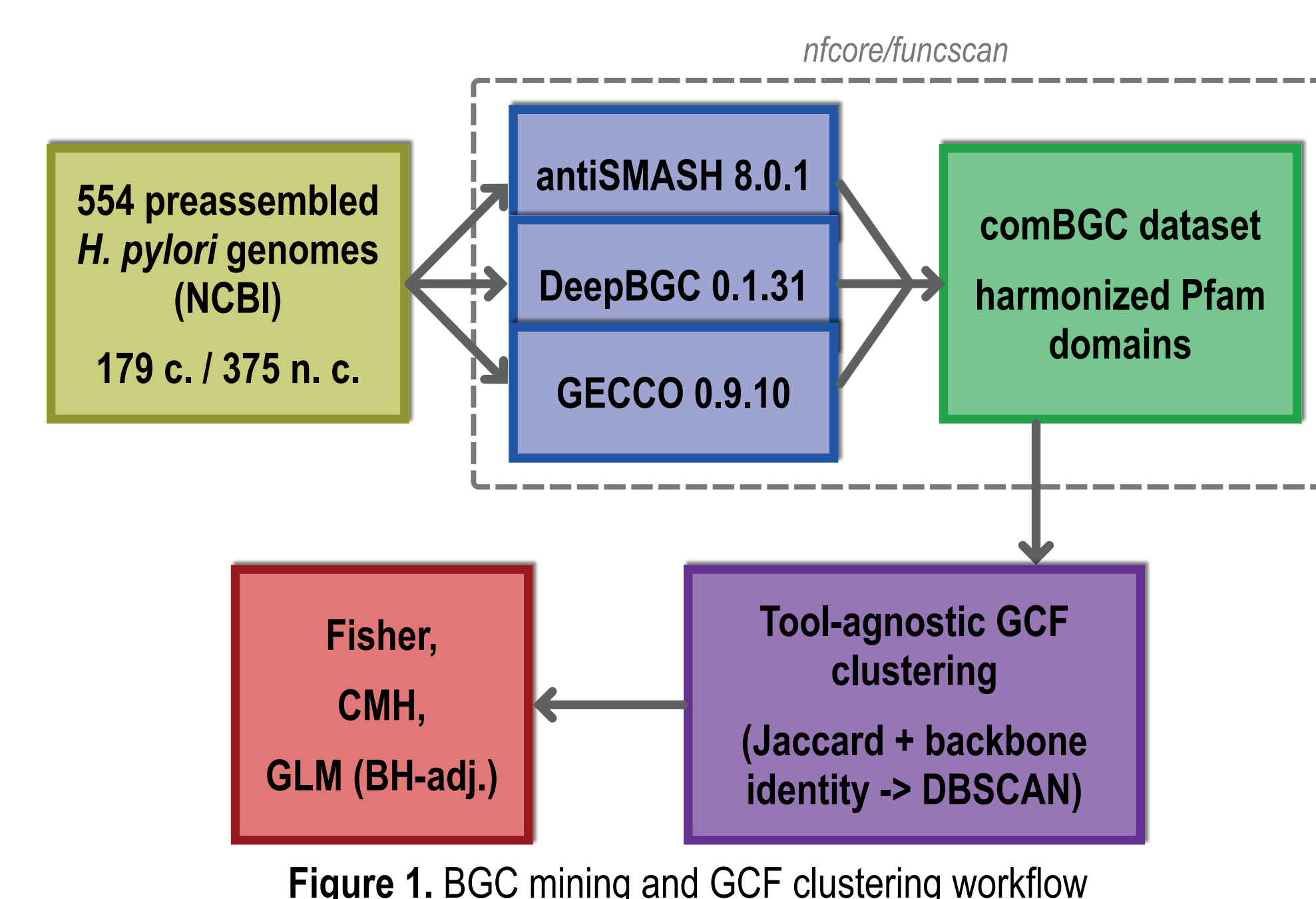


Figure 1. BGC mining and GCF clustering workflow

- BGC mining: nf-core/funcscan (antiSMASH + DeepBGC + GECCO); results harmonized into Pfam domain arrays by comBGC.
- GCF clustering: tool-agnostic method. For backbone-verified classes (43.1% of qualifying pairs):

$$\text{Similarity} = \sqrt{(0.8 \times \text{hmmalign sequence identity} + 0.2 \times \text{Jaccard})}$$

$\sqrt{(0.2 \times \text{Jaccard})}$ otherwise. DBSCAN ($\epsilon = 0.56$, $\text{min_samples} = 2$).

- Statistics: Fisher's exact test, Cochran–Mantel–Haenszel (stratified by continent), multivariable logistic regression (continent + age + BGC richness); Benjamini–Hochberg correction across the full qualifying set of GCFs (≥ 3 carrier genomes, decided a priori).

Results

- 2,076 BGC predictions → 1,828 (88.1%) grouped into 135 GCFs, 248 remained as singletons.
- Dominant GCF (554 BGCs) = universal terpene-precursor locus, present in all genomes. Remaining GCFs' median size = 3.
- BGC richness per genome: higher in cancer (median 4 vs. 3; Wilcoxon $p = 5.96 \times 10^{-5}$) — driven by DeepBGC ($p_{\text{adj}} = 2.6 \times 10^{-5}$).

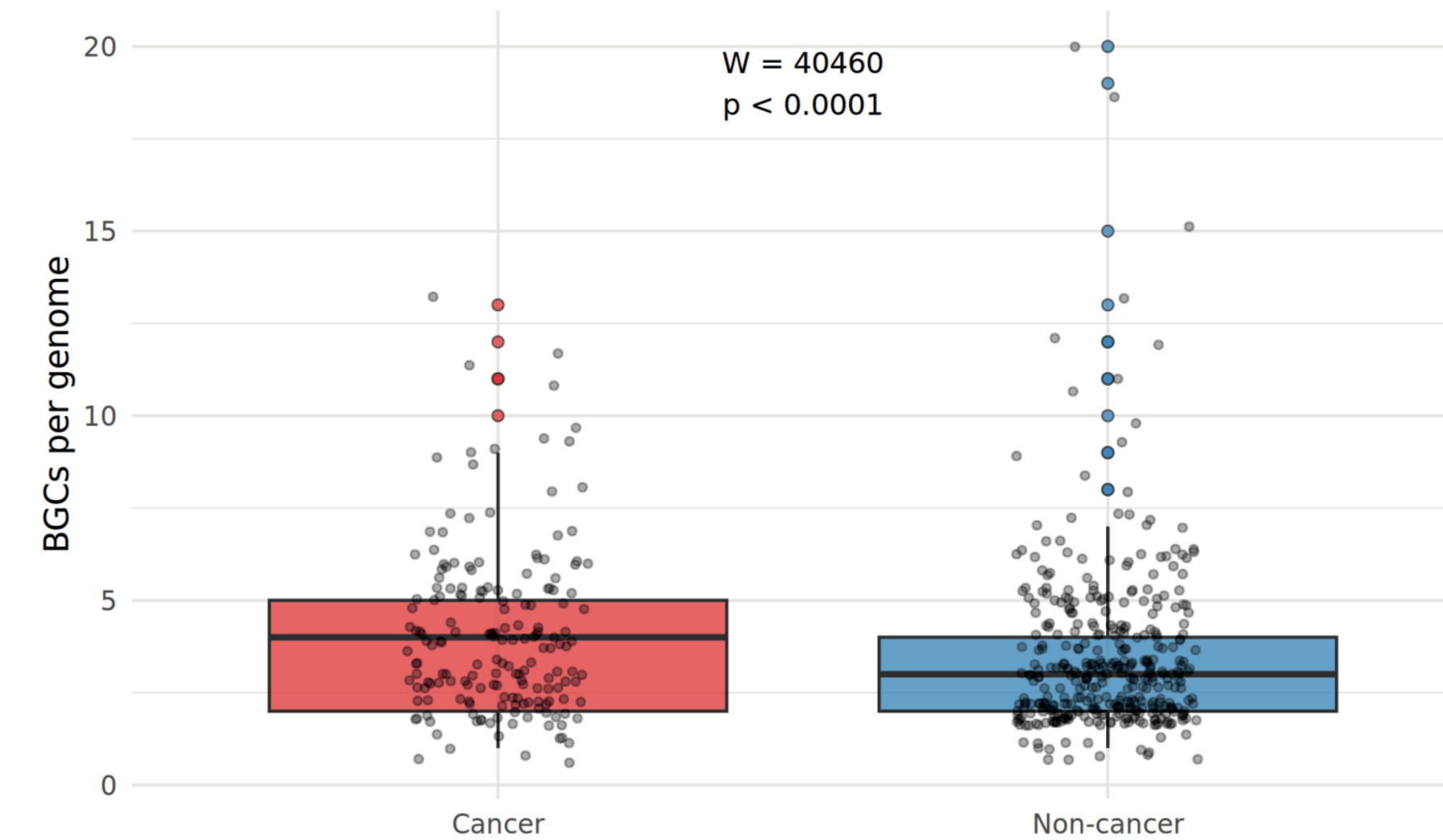


Figure 2. BGC richness by clinical group

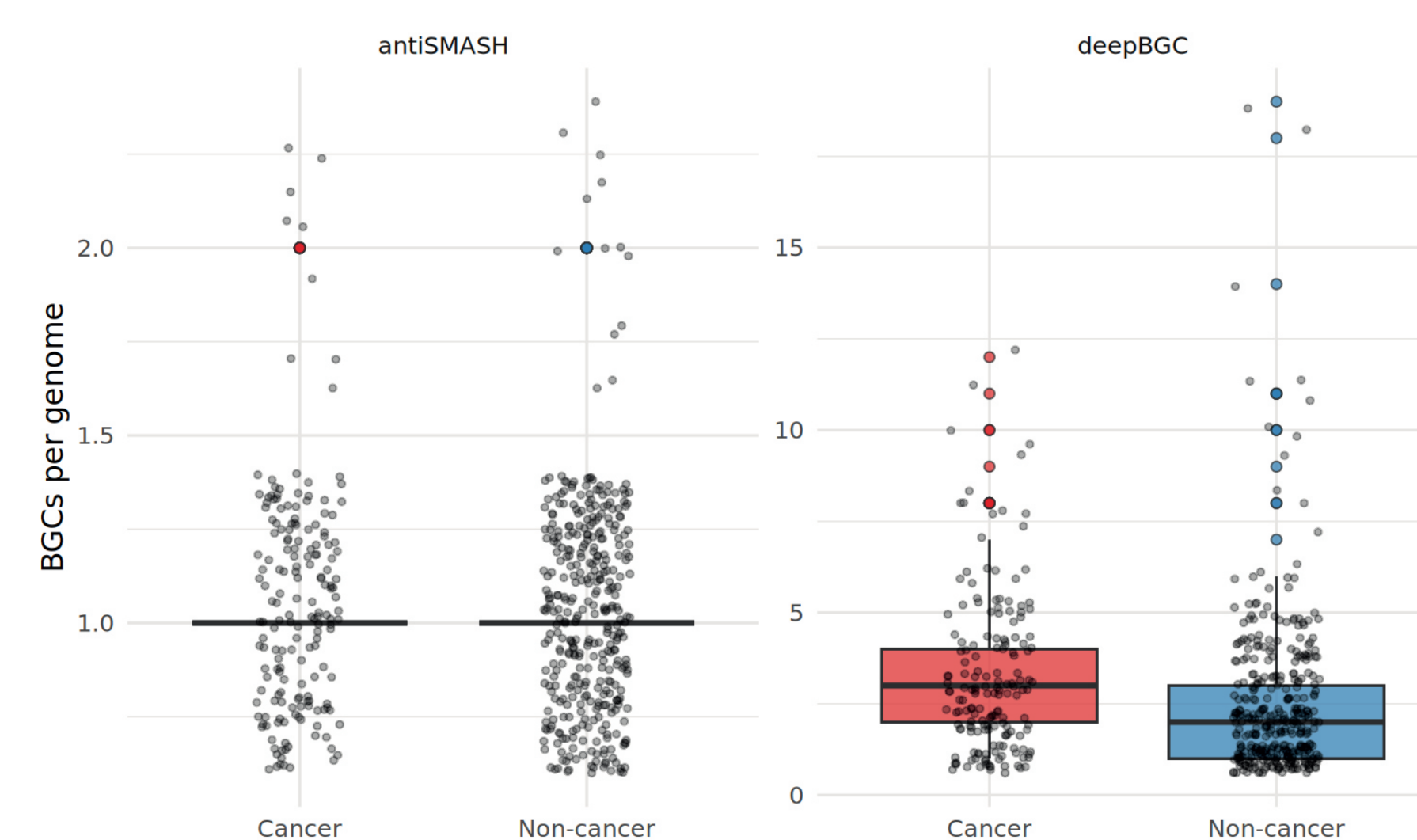


Figure 3. BGC richness by clinical group and mining tool

- Curiously, despite having equal median, GCF richness also differs (Wilcoxon $p = 2.1 \times 10^{-4}$).

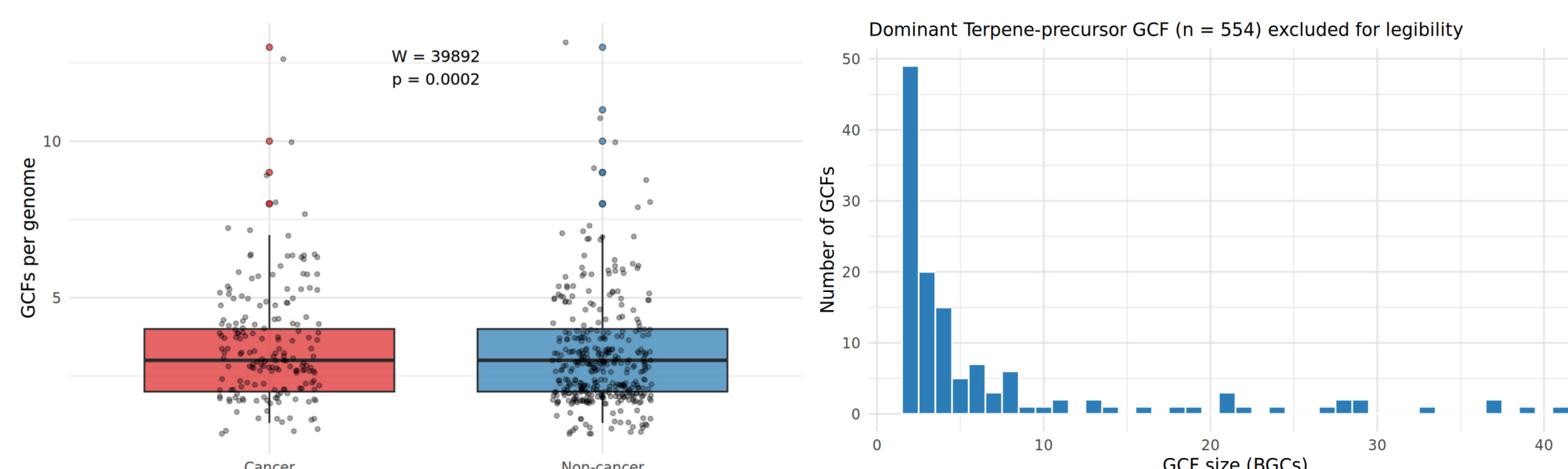


Figure 4. GCF richness by clinical group. Figure 5. GCF size distribution (≤ 50 BGCs)

- No individual GCF reaches significance: of 85 GCFs meeting the ≥ 3 -genome criterion, no Fisher CMH or logistic hits.
- Quasi-Poisson model: cancer-phenotype IRR = 1.119 (95% CI 1.00–1.25, $p = 0.049$, borderline) independent of age (IRR = 1.005/year, $p = 0.0047$); no age $\times$ phenotype interaction ($p = 0.864$).

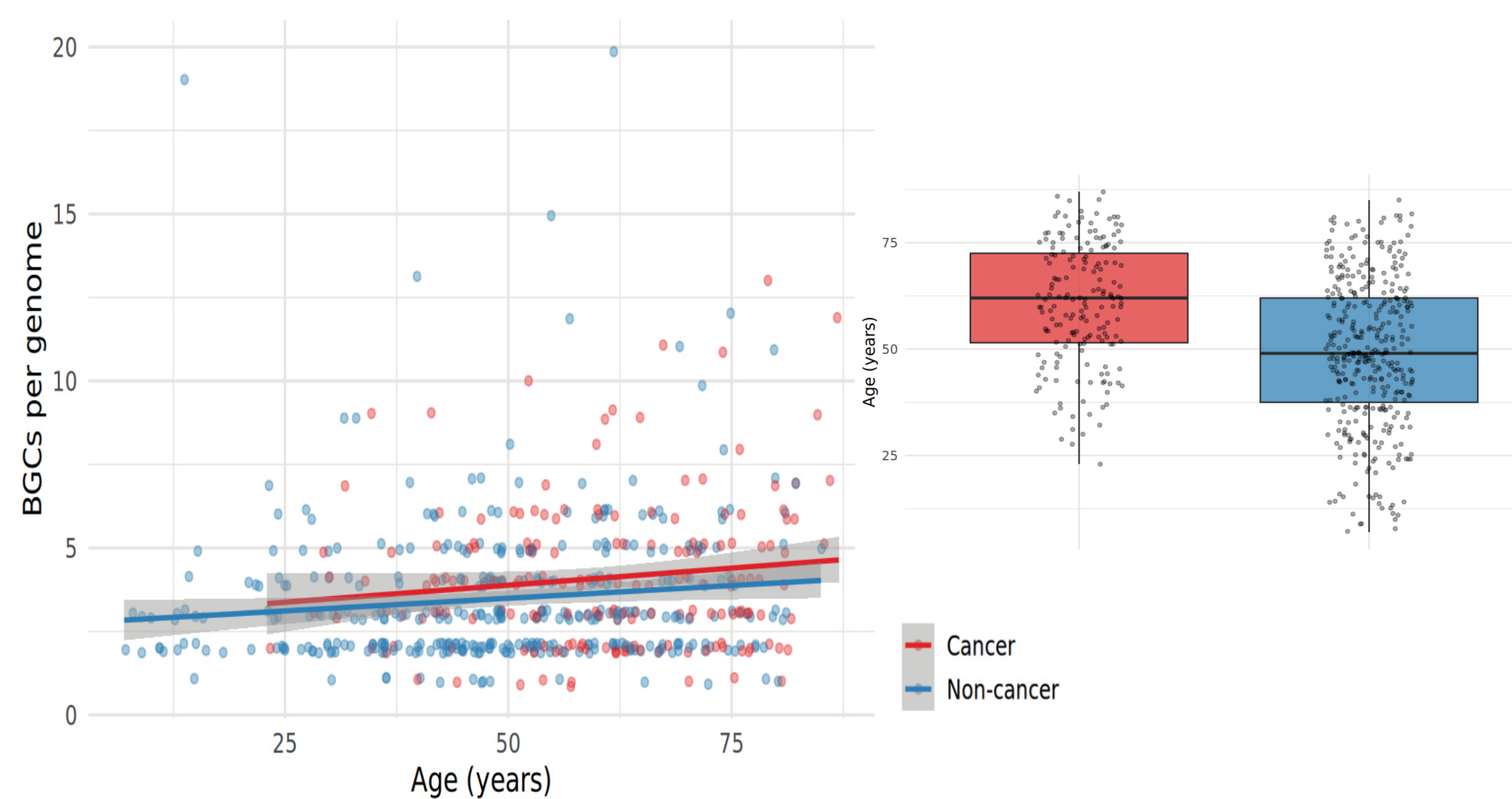


Figure 6. BGC richness vs age, by clinical group. The boxplot displays the age distribution of the cohort, which justifies the interaction analysis.

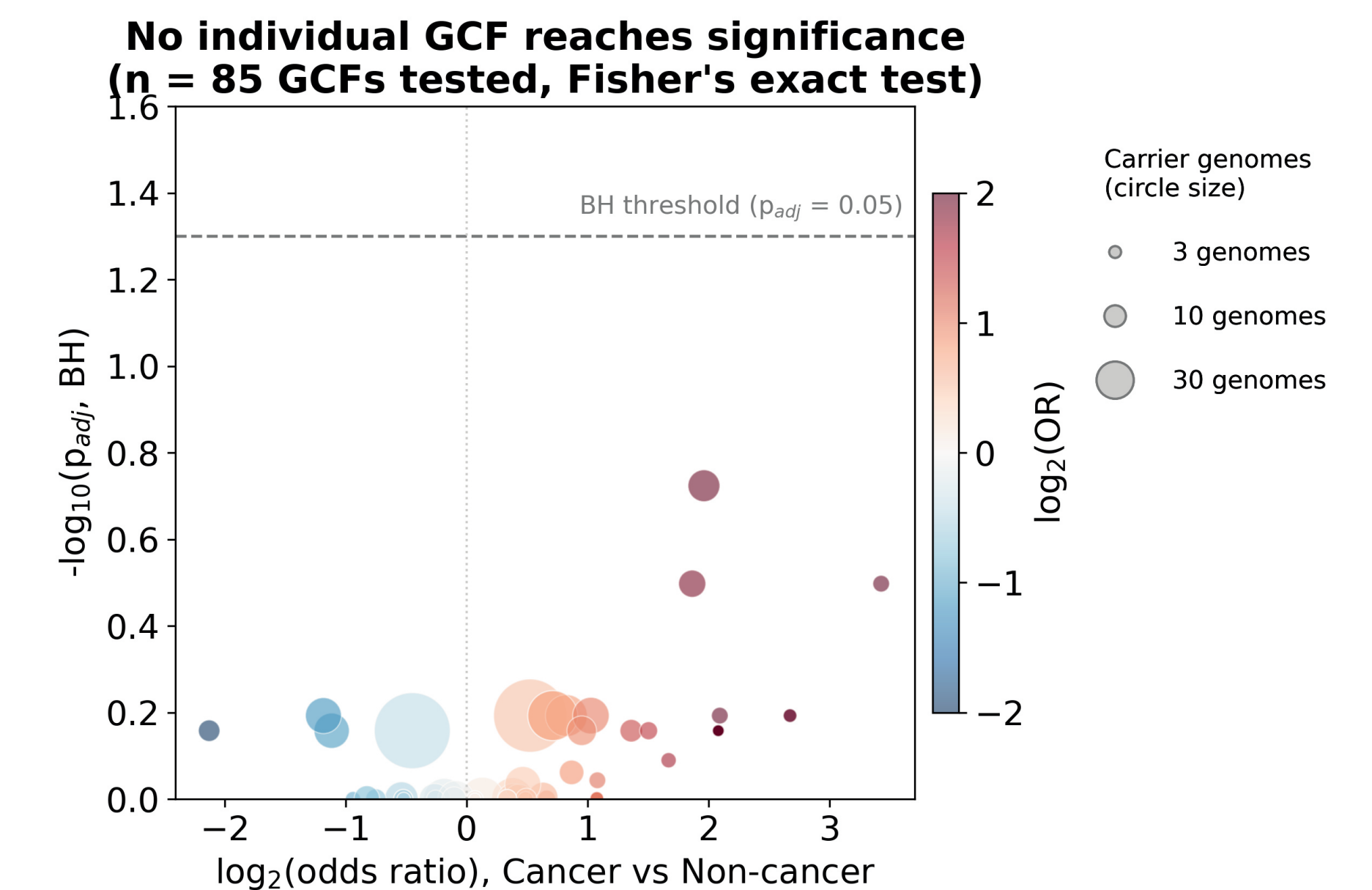


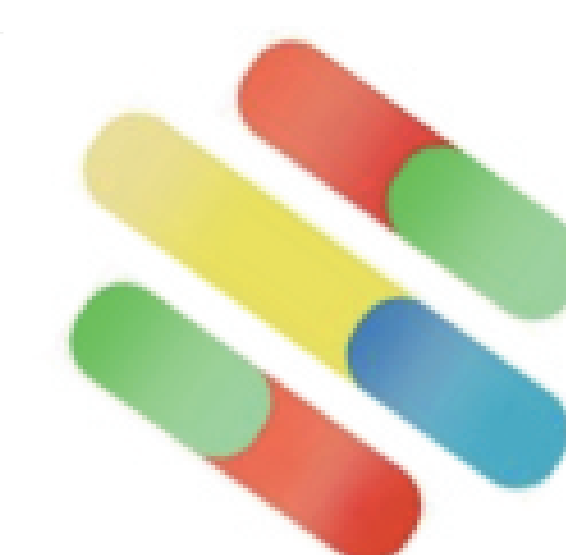
Figure 7. Volcano plot representing individual GCFs' significance towards cancer phenotype (Fisher's exact test).

Discussion

- Taken together, the results reveal a subtle but statistically robust increase in overall BGC richness in cancer genomes, but with no individual GCF surviving correction. Evidence hints that **phenotype-related effects may manifest more as quantitative differences in overall biosynthetic load than as the presence or absence of specific families** — consistent with *H. pylori*'s restricted, highly conserved biosynthetic repertoire (echoing the pilot study).
- The breakdown by prediction tool highlights the value of combining rule-based and deep-learning genome mining for BGC discovery.
- GCF structure was mainly driven by hmmalign-based identity scoring: backbone-domain identity is the dominant signal for GCF granularity here.
- Limitations: cohort imbalance (179 cancer vs. 375 non-cancer); the cancer cohort also skews older individuals, though the quasi-Poisson model shows age and phenotype act additively, not confounding each other; residual confounding beyond continent/age (strain background, finer geography, treatment history not in the metadata) remains possible; the tool-agnostic GCF method is new and not yet benchmarked against BiG-SCAPE.

Next steps

- Correlate *H. pylori* phylogeny with the observed BGC repertoire.
- Search for a malignancy signature beyond univariate testing.
- Extend backbone-domain identity resolution to GECCO (37 BGCs) and DeepBGC's remaining low-specificity classes.
- Discuss an experimental approach for functional validation of the most relevant BGCs in an animal model.



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